APETIT × SINCE AI HACKATHON 2026

Climate & Nature Navigator

From raw, error-ridden data to approved climate and nature decisions — every number traceable to its source.

Built on the synthetic Since AI package v1.0.0 · results illustrate the method, not Apetit's actual emissions

The problem: fragmented data gives confidently wrong numbers

314×

how far the draft 2025 report overstates Scope 3: **71.5 Mt** instead of **228 kt**

The draft was typed straight from the uncleaned pipeline:

- kWh read as MWh on meters
- tonnes keyed as kg in the ERP
- refrigerant grams entered as kg
- a group total that does not add up

And the farm calculator assumes every field is mineral soil — the biggest real emission source is invisible.

Source: data/05_scenarios/draft_report_summary_2025.csv vs cleaned inventory (A/lines_long.csv)

One pipeline, five steps, a human sign-off at the end

1 · BRONZE

Raw feeds

26,474 records from ERP, meters, fuels, GoOs, surveys, PCFs

2 · AGENT

Data-quality agent

18 detectors; confidence, explanation, proposed fix

3 · SILVER

Clean tables

every correction logged with old → new value and reason

4 · GOLD

Inventory & models

GHG Protocol, SBTi + FLAG, NGFS/SSP, SBTN

5 · APPROVE

Expert review

draft → approved per module, exportable audit log

Delivered as one working app: decision brief · DQ issue queue · inventory drill-down · SBTi pathways & MACC · climate risk · catchment maps · copilot that cites its evidence.

Reproduce: python run_all.py <package> — about 2 minutes, byte-identical outputs

It works: checked against every public answer key

Task	Ours	Package baseline	Checked on
A1 data-quality issues	**96.0**	52.2	2023–24 labels
A2 GHG inventory 2023 / 2024	**95.9 / 96.6**	25.5 / 23.2	published truths
C1 yield forecast (production error)	**6.8 %**	11.6 %	leave-year-out 2014–26
D1 catchment loads & ranking	**76.7**	12.0	6 dev catchments (LOO 73.5)
E copilot, numeric dev	**3 / 3**	–	dev answers

Recall is 100 % on 12 of 13 issue types; the only miss is organic soils, where the public soil map is itself ~88 % accurate — so those cases go to an expert.

Finding 1 · Organic soils are the hidden majority of farm emissions

59 %

of Finnish contract-farm emissions come from drained organic soils — 15,774 t the current calculator leaves out

6×

pea footprint is 1.04 kg CO2e/kg, not 0.17; turnip rape 2.43, not 0.87

27

farms have ≥ 30 % of contract area on organic soil; 17 "mineral soil" answers routed to the farmer for confirmation

Decision: replace the mineral-soil default with parcel soil data. It also opens the cheapest big lever: moving contract area off peat (L11) ≈ 7,900 t/yr at ~€11/t, with water-quality co-benefits.

Source: A/farm_footprints_2025.csv, A/organic_soil_impact.csv; METHODOLOGY §5 (29 t CO2/ha/yr)

Finding 2 · SBTi: a 2025 base with FLAG, and where the gap closes

Base year 2025, adjusted

Full year of Foodhills added (Bjuv gas + 298 Swedish pea growers). FLAG = **70 %** of the total, so a FLAG target is mandatory. A 2019 base would need grain-trade estimates and has no FLAG split.

2030 targets

Scope 1+2: 5,819 → 4,597 t (−4.2 %/yr)
Scope 3 non-FLAG: 67,857 → 59,375 t (−2.5 %/yr)
FLAG: 164,299 → 139,408 t (−3.03 %/yr)

Cheapest big levers

Rapeseed supplier PCFs + low-carbon fertiliser + origins (L13/L14): ~22 kt/yr at €31–52/t
Off-peat (L11): 7.9 kt at €11/t
Biogas + R404A (L01/L07/L03): meets S1+2

Tested under 7 NGFS scenarios. Structural moves (Pudasjärvi, Kantvik steam) and better data (L27) are shown — and **never counted as reductions**. Our simulator shows FLAG misses the 2035 line by 17–22 kt with today's lever catalogue. Closing it means roughly doubling the off-peat and rapeseed programmes: about 84 kt/yr at an average of €39/t.

Source: B/b_results.json, B/pathways.csv, B/levers_recomputed.csv · carbon prices are illustrative NGFS-range values

Finding 3 · Weather risk is already here, and two countries hedge it

1 in 7

years the Finnish pea crop falls below 80 % of normal (turnip rape: 1 in 13). Expected shortfall cost ≈ **€0.46 m/yr**, ~€1.5 m in a 1-in-10 year, plus ~2 kt CO2e from imports.

−72 %

expected pea shortfall cost when sourcing from Finland and Sweden instead of Finland only (SSP2-4.5); −44 % already today. 2026: FI +24 %, SE −8.5 %.

+€350k

net per year from irrigating potato and carrot parcels; heat-tolerant varieties pay back for pea, spinach, carrot, rape. Irrigating Swedish peas does not.

To 2041–60 the shift is modest (SSP5-8.5: SE pea bad years 4.7 → 6.5 %, potato shortfall cost +31 %), measured with a 40,000-draw Monte Carlo on yield models with 6.8 % backtest error. The decision is to hedge now, not to wait for 2050.

Source: C/c3_euro_risk.csv, submissions/C2_*.csv, C/c3_hedge_fi_se.csv, C/c3_adaptation.csv · irrigation cost assumed €250/ha/yr

Finding 4 · Nature: 50 fields carry 15 % of the phosphorus

Contract parcels deliver **7.6 t P** and **91 t N** to water in 2025, modelled field by field: soil-P, clay, slope, crop, rain, buffer strips, catch crops, distance to water, retention.

Priority = pressure × state of nature (SBTN step 2). Top catchments: **CA12, CA24, CA14, CS07, CA06**.

Value-chain screen: soy protein isolate (CN) ranks first on nature risk — outside EUDR (HS 3504) but an SBTN high-impact commodity.

Measure on hotspots	**€/kg P**
Stop P on high soil-P fields	38–70
Gypsum on clay	38–70
Buffer strips	150–250
Catch crops	150–250
Top-5 catchment programme	**−43–52 % P**

Source: D/hotspots_top50.csv, D/catchment_programme_top5.csv, D/d2_materiality.csv · measure costs indicative

Why Apetit should buy it

Right numbers, first time

Catches a 314× misstatement before an auditor or SBTi validator does.

Finds what's missing

15.8 kt of peat-soil emissions the current calculator cannot see.

SBTi-ready by Dec 2027

Base year, FLAG split, targets and a costed plan: 37 kt/yr available below €60/t.

Euros on climate risk

€0.46 m/yr supply risk priced per crop; the hedge and irrigation that cut it.

Nature where it matters

50 fields, 15 % of phosphorus; fixes from €38/kg P. Ready for SBTN and TNFD.

AI that cites, not guesses

A tool-using copilot and board-memo writer, every answer backed by evidence.

One data model replaces spreadsheets and one-off consultant studies, and updates every time the data does. Every number is traceable, nothing is published without a named expert's approval, and it runs on Apetit's Microsoft Fabric and AI Foundry stack.

Why Apetit can trust it: the agent proposes, the expert approves

Every number has a source

Hover any figure: record keys, factor id and its source. 190 inventory line items, each with its method.

Every correction is logged

759 findings with old → new value, rule, evidence and confidence; 223 routed to an expert. Source data is never overwritten.

Nothing publishes in draft

Each module needs a named reviewer's approval; decisions export as an audit log.

The copilot cites, never guesses

Answers come from retrieved evidence and the pipeline's numbers; with no evidence it says so.

What Apetit should do next week

Action	Owner	Size
Block the draft 2025 summary; publish the cleaned inventory	Reporting	avoids a 314× misstatement
Use parcel soil data in the farm calculator	Sustainability + field team	+15.8 kt correctly counted
Set SBTi base 2025 (adjusted) with a FLAG target	Head of sustainability	submit by Dec 2027
Rapeseed supplier programme (PCFs, low-carbon N, origins)	Procurement	~22 kt/yr at €31–52/t
Biogas at Säkylä and Bjuv, replace R404A	Operations	meets Scope 1+2 2030
Nutrient programme in 5 priority catchments	Agronomists	−43–52 % contract P
Keep FI + SE pea sourcing; irrigate potato and carrot	Contract farming	−72 % shortfall cost, +€350k/yr

Path to production on Apetit's Microsoft stack

Layer	Today (hackathon)	Apetit production
Data	CSV / Parquet	Fabric Lakehouse: bronze → silver → gold (Delta)
DQ agent	Python detectors	Fabric notebooks + Azure AI Foundry agent
Models	LightGBM, nutrient model	Azure ML / Fabric Data Science, MLflow
Copilot	in-browser retrieval + optional LLM	Azure AI Search + Azure OpenAI, Copilot Studio
Approval	in-app sign-off, JSON audit log	Power Apps flow, log as Delta table, Power BI
Geospatial	synthetic GeoJSON	SYKE catchments, Ruokavirasto parcels, Azure Maps

Not just calculating the past — deciding the future, with an audit trail.

Working app · reproducible code (one command, ~2 min) · submissions for A1, A2, C1, C2, D1 and E · method note.

Live demo + code: see the GitHub repository · apetit-climate-nature-navigator